# Trust Index Notes

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## Abstract

Working notes on the definition of AI-judge trustworthiness.

## 1 Notation

Table 1: Notation.

| Symbol | Meaning |
|---|---|
| Items, agents, and benchmarks | |
| $x$ | candidate artifact |
| $y^*$ | gold / reference artifact |
| $\hat{y},\ \hat{y}(c)$ | agent output on input $c$ |
| $A$ | set of agents / solvers |
| $a \in A$ | an agent / solver |
| $B$ | set of benchmarks |
| $b \in B$ | a benchmark |
| Judges and targets | |
| $AJ$ | set of AI judge configurations |
| $aj \in AJ$ | AI judge configuration (model, prompt, $K_{aj}$, aggregation) |
| $TJ$ | set of target judges |
| $tj \in TJ$ | a target judge |
| $\theta_j,\ \theta_{aj}$ | parameters of $aj$ (prompt, weights, feedback, …) |
| $h_j,\ H$ | human / target judge |
| $t_j$ | target persona (or the target itself) |
| $a_j$ | persona realized by configuration $aj$ |
| $\pi,\ \Pi$ | rater persona; set of personas |
| $K_{aj}$ | number of judge repeats |
| $K_a$ | number of agent generations |
| $K_h$ | number of human passes |
| Scores and distributions | |
| $s_{aj}(x, y^*, \hat{y})$ | score of $aj$ on a triple; range $[0, 1]$ |
| $\hat{s}_{aj}$ | finite-sample estimate of $s_{aj}$ |
| $s^*_T,\ \hat{s}^*_T$ | target score and its finite-sample version |
| $p_{AJ}(s \mid \cdot)$ | score distribution (pdf/pmf) of the AI judge |
| $p_T(s \mid \cdot)$ | score distribution (pdf/pmf) of the target judge |
| $\hat{p}_{AJ},\ \hat{p}_T$ | finite-sample versions of the above |
| Trust objects | |
| $\mathrm{TW}_m(aj)$ | trustworthiness of $aj$ at matching level $m$ (construct) |
| $\mathrm{TI}_m(aj)$ | Trust Index of $aj$ at matching level $m$ (criterion) |
| $\mathrm{PTI}_p(aj)$ | Proxy Trust Index of $aj$ on property set $p$ (instrument) |
| $m,\ m_0, m_1, m_2$ | matching-criterion index and its ladder |
| $\approx_\Pi$ | persona precondition ($t_j = a_j$) |
| $\approx_m$ | match under criterion $m$ |

Notation note. $\hat{p}_T$ denotes the target-judge pdf/pmf; the hat indicates that the estimate is finite-sample. We write $\hat{p}_T$ (or $\hat{p}_H$ when the target is human) and $\hat{p}_{AJ}$ for the AI-judge side.

## 2 Behaviors and agents

Definition 1 (Universally trustworthy AI judge). Let $B$ be a set of benchmarks and $A$ a set of agents. An AI judge $aj \in AJ$ is universally trustworthy if, for every benchmark $b \in B$ and every agent/solver $a \in A$, $aj$ distributionally matches the corresponding target judge ($tj \in TJ$) (e.g., a human, a persona, a strict rubric, a lenient rubric, etc.):

$$\forall b \in B,\ \forall a \in A, \quad \text{if } t_j = a_j \quad \Rightarrow \quad \hat{p}_{AJ}\big(s \mid x,\, \hat{y}(c),\, y^*,\, \theta_j\big) \;=\; \hat{p}_T\big(s \mid x,\, \hat{y}(c),\, y^*,\, \theta_* + t_j\big),$$

where equality may be measured by KL divergence (or total variation).

Remark. The precondition $t_j = a_j$ (persona / target match) is required. If, for example, we set $t_j =$ strict but the judge $a_j$ has parameters/configuration $\theta_{a_j}$ (prompt, weights, etc.) that map to a lenient judge, then the definition is unsatisfiable.

Remark (preference vs. trustworthiness). In the current literature, "trustworthiness" of an AI judge is almost always operationalized as matching human judgments (i.e., tracking human scores). "Preference" is the more common term in RLHF settings (which answer is preferred). The two notions are close once a frame of reference is fixed, but they are not identical: one can trust a system whose outputs one does not personally prefer, and vice versa.

## 3 Practical instantiations

Definition 2 (Common practice ($m_1$)). Most work treats an AI judge as trustworthy when its scores correlate with human scores on a panel:

$$\mathrm{TI}_{m_1}(aj) \;=\; \mathrm{Corr}\big(\hat{s}_{aj},\, \hat{s}_T\big).$$

Remark (what Corr actually matches). Correlation is a property of the joint distribution of the paired scores $(\hat{s}_{aj}(i), \hat{s}_T(i))$. It is a dependence / association measure, not a pure marginal moment match and not a full distributional match ($m_0$).

- Pearson is standardized covariance (a joint second-moment quantity after centering and scaling). It can be 1 even when the means differ (by affine rescaling).
- Spearman and Kendall match ranks / pairwise orderings of items; they are still weaker than full distributional equality.

A high correlation therefore says that the judge ranks items similarly to the target; it does not say that the two score distributions are close, nor that their means match.

The stability term $\mathrm{std}(\hat{s}_{aj})$ (used in $\mathrm{TI}_{2'}$) is a pure marginal property of the judge alone and need not match the human at all.

Definition 3 (Valid-and-stable ($m_2$ / $\mathrm{TI}_{2'}$)). A stricter practical criterion combines validity (correlation) with stability (low repeat variance) via a geometric mean:

$$\mathrm{TI}_{2'}(aj) \;=\; \Big(\mathrm{Corr}\big(\hat{s}_{aj},\, \hat{s}_T\big) \cdot \big(1 - \mathrm{std}(\hat{s}_{aj})\big)\Big)^{1/2}.$$

This is the working definition of trustworthiness$_{2'}$ used in these notes.

Remark (Sanmi's objection to the stability term). The term $1 - \mathrm{std}(\hat{s}_{aj})$ is a pure marginal property of the judge. It does not attempt to match the human's own variability. Consequently a judge that is artificially more stable (or less stable) than the humans can still score highly, provided the ranking agreement remains strong. This is a deliberate design choice (penalize unreliable judges) rather than a pure "match the human" criterion. Full distributional match of both location and scale would be more faithful but is currently impractical. Natural alternatives for later work include:

- also matching $\mathrm{std}(\hat{s}_{aj}) \approx \mathrm{std}(\hat{s}_T)$,

- using a concordance measure (e.g. Lin's CCC) that penalizes both location and scale differences,
- or reporting correlation and judge stability as two separate numbers instead of a single geometric mean.

Remark (ideal vs. practical). The ideal target is full distributional match of scores once a persona is fixed (Definition 1). In practice this is expensive: many humans and many repeated scores would be required. We therefore approximate the ideal by an association of collapsed scores (correlation) plus a judge-side second moment (low repeat std), combined by geometric mean to obtain $\mathrm{TI}_{2'}$. The practical criterion only captures limited aspects of the ideal match—and the std conjunct is not a match to humans at all.